

Project Design Phase

Proposed Solution Template

Date	7 July 2026
Team ID	SWTID-2026-2886
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Processing, cleaning, and analyzing massive, unorganized power grid datasets (16,800+ lines of time-series data) across 5 national zones of India is highly inefficient. Stakeholders lack optimized visual tools to easily isolate past macro-event anomalies (like the 2020 national lockdown drops) against historical 2019 baseline data without facing query lag or calculation glitches.
2.	Idea / Solution description	An interactive, high-performance visual analytics dashboard built using a decoupled architecture. The solution utilizes backend Python (Pandas/NumPy) scripts for rapid data transformation and cleanup, paired with a frontend Tableau/Power BI interface. It incorporates custom Context Filters and synchronized dual-year timeline maps to provide smooth, instant cross-filtering and sub-regional data analysis.
3.	Novelty / Uniqueness	Unlike standard static reports, this solution uses specialized Context Filtering layers that fix the data calculation order. This completely eliminates dynamic "Top-N" state ranking errors when changing regional views. Furthermore, it incorporates in-memory data extracts to allow heavy time-series calculations to render in under 2 seconds.
4.	Social Impact / Customer Satisfaction	Social Impact: Enables national energy boards to better understand load behavior during macro-crises, leading to improved grid stability and reduced blackouts. Customer Satisfaction: Power grid analysts can extract clear, accurate insights and export reporting sheets within just 3 intuitive clicks, removing database maintenance headache
5.	Business Model (Revenue Model)	As an internal analytical infrastructure project, the primary returns are Operational Cost Savings through enhanced resource allocation and minimized power distribution losses. Alternatively, it can follow a B2G (Business-to-Government) Software-as-a-Service (SaaS) model, providing premium custom dashboard access and automated reporting subscriptions to state electricity boards.
6.	Scalability of the Solution	The architecture is highly scalable due to its modular design. By isolating the backend processing layer from the frontend reporting tool, the pipeline can smoothly expand to ingest new multi-year datasets (e.g., tracking consumption patterns from 2021 to 2026) or connect with real-time smart meter APIs without crashing the core dashboard.